How accurate is altitude from pressure geolocators? A global assessment of three pressure-to-altitude conversions on the ground and in flight

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Abstract

Background: Pressure geolocators record atmospheric pressure throughout the year on small birds, and pressure sensors also provide vertical records in GPS-equipped tags. The altitudes derived from these records reveal migration strategies and inform conservation assessments, including exposure to wind turbines. However, pressure-to-altitude conversions make assumptions about atmospheric conditions whose consequences for altitude error remain poorly quantified.

Methods: We evaluated three conversions: a fixed standard atmosphere, a conversion using local ERA5 surface conditions, and an ERA5 conversion additionally accounting for virtual temperature and a climatological temperature lapse rate from GPT3. Treating reference pressures as exact conversion inputs, we compared estimates with 2,130 GHCNh weather stations worldwide and IGRA2 radiosondes from 643 stations up to 6 km above ground. We summarize absolute errors by their medians and 95th percentiles, distinguish station bias from within-station SD, and weight flight-height errors by the flight time of 203 tracked birds.

Results: At screened ground-reference stations, the standard atmosphere produced a median absolute error of 40.3 m, with 5% of absolute errors exceeding

148 m. Local ERA5 conditions reduced these to 4.1 m and 23 m, respectively. In flight, flight-time-weighted median absolute errors were 20.6 m with the standard atmosphere, 6.9 m with ERA5, and 5.5 m after adding virtual temperature and GPT3. At 2–3 km, the correction reduced the median station bias from -33.2 m to -5.5 m. At 2–3 km above ground, the 95th percentile of absolute flight-height error fell from 237 m with the standard atmosphere to 84 m with the corrected ERA5 conversion.

Conclusions: Local atmospheric conditions substantially reduce the large altitude errors introduced by a fixed standard atmosphere; virtual temperature and a climatological lapse rate further reduce systematic flight-height errors. The conversion is computationally simple and available in a software function using measured pressure, measurement time and an estimated geographic position. Remaining atmospheric and sensor errors must still be considered when interpreting flight behaviour or classifying flights relative to turbine heights.

Keywords: biologging, barometric altimetry, flight altitude, ERA5, radiosonde, multi-sensor geolocator, GeoPressureR

1 Background

Altitude is a central dimension of how birds use their environment. In flight, it helps explain how migrants respond to winds and temperature [1, 2], how birds partition aerial foraging space [3], and how they use soaring flight [4, 5]. Vertical records also reveal behaviours whose functions remain incompletely understood, including twilight ascents in swifts [6], exploratory climbs during migration [7], and pronounced day–night changes in flight altitude [8, 9]. On the ground, elevation describes habitat use and movements such as elevational commuting, uphill roosting and seasonal migration [10–12]. Height distributions also inform conservation assessments, including exposure to wind turbines [13, 14].

Miniature pressure sensors make it possible to study these vertical movements over long deployments, including in small migratory birds carrying multi-sensor geolocators [15–17]. Pressure sensors are also incorporated into GPS-equipped tags, extending their applications to larger birds [4, 5, 13]. Alongside these biological applications, pressure-derived elevation can help constrain geographic positions during track reconstruction [18]. Interpreting these records requires knowing how accurately pressure can be converted into altitude, and whether the resulting uncertainty is small enough to resolve the movements or height differences of interest [19].

Pressure-to-altitude conversion requires a reference pressure at a known altitude and an estimate of the temperature profile between that level and the bird. The simplest approach uses a fixed standard atmosphere [20], as in several geolocator studies [7, 17, 21, 22]. Other studies obtain reference conditions from local weather stations [3], reanalysis sea-level pressure [6], or periods when birds are assumed to be at the sea

surface [13, 23]. Reanalysis can also supply the temperature information needed for conversion: GeoPressureR uses ERA5 surface pressure, temperature and orography at an estimated position [18, 24, 25], while Schaub et al. [26] additionally estimated the temperature lapse rate from surface and upper-air conditions.

Previous evaluations have established the performance of pressure-derived heights in controlled device trials [27], at grounded birds [26], and through comparisons with GPS during flight [23]. These studies provide important evidence about particular devices, conversion methods and field conditions. However, their results do not establish how atmospheric conversion errors vary across the climates, terrain and flight heights encountered during long tracking deployments. Quantifying this component separately helps explain when errors arise from the atmospheric assumptions and how much changing those assumptions can improve altitude estimates.

Here, we compare three pressure-to-altitude conversions: a fixed standard atmosphere, an ERA5 surface reference with the standard lapse rate, and the same reference with virtual temperature and a climatological lapse rate from GPT3 [28]. Using worldwide weather stations, stations with documented pressure-sensor heights and radiosonde profiles up to 6 km above the ground, we quantify atmospheric conversion errors on the ground and in flight. We distinguish systematic bias from temporal variability, characterize the range of errors, and summarize flight-height errors using the heights of tracked birds. Our aim is to establish how much local atmospheric information and a more realistic temperature profile improve a conversion suitable for long pressure records.

2 Methods

2.1 Three conversions from pressure to altitude

Pressure decreases with height because each layer of air supports the weight of the air above it. Combining this hydrostatic approximation with the ideal gas law gives the hypsometric relation [29]:

$$z = z_0 + \frac{R\,\bar{T}_v}{g\,M}\,\ln\frac{p_0}{p}, \tag{1}$$

with z the altitude (m) at pressure p, z_0 and p_0 the altitude and pressure of a reference level, $g = 9.80665$ m s^{-2}, $R = 8.31432$ J mol^{-1} K^{-1}, $M = 0.0289644$ kg mol^{-1}, and $\bar{T}_v$ the mean virtual temperature.

The virtual temperature $T_v = T(1+0.608\,q)$, with T the air temperature in kelvin and q the specific humidity, accounts for moist air being lighter than dry air. Temperature can vary with height; the conversion uses the mean virtual temperature between the reference level and the bird, weighted by increments of log pressure:

$$\bar{T}_v = \frac{1}{\ln(p_0/p)}\int_{p}^{p_0} T_v(p')\,d\ln p'.$$

To simplify the computation, temperature is assumed to change at a constant rate with height, giving a linear profile $T(z) = T_0 + L(z - z_0)$, where T_0 is the reference temperature in kelvin. This rate, called the lapse rate $L = dT/dz$ (K m^{-1}), is negative when temperature decreases with height. Without a humidity correction, $T_v = T$, and integration gives the barometric equation:

$$z = z_0 + \frac{T_0}{L}\left[\left(\frac{p}{p_0}\right)^{-RL/(gM)} - 1\right]. \tag{2}$$

The three conversions differ in their reference level and in the column temperature they assume (Figure 1):

1. **Standard atmosphere.** The reference is sea level ($z_0 = 0$, $p_0 = 1013.25$ hPa) with $T_0 = 288.15$ K (15 °C) and $L = -6.5$ K km^{-1} [20]. Its main advantage is that it needs no external data, but these fixed conditions do not account for changes in pressure, temperature or humidity.
2. **ERA5, standard lapse rate.** The ERA5 reanalysis provides surface pressure p_0, 2 m air temperature T_0 and surface geopotential Φ at the location and hour of each measurement, with reference altitude $z_0 = \Phi/g$. This accounts for local pressure and temperature variations, while retaining a constant lapse rate of -6.5 K km^{-1} and no humidity correction.
3. **ERA5, virtual temperature and GPT3 lapse rate.** This conversion adds two refinements to the ERA5 reference. First, humidity at the reference level is accounted for by replacing T_0 in Equation 2 with the reference-level value of T_v. The specific humidity q needed for this correction is calculated from ERA5 surface pressure p_0 and 2 m dewpoint T_d, via vapour pressure e [30]:

$$q = \frac{0.622\,e}{p_0 - 0.378\,e}, \qquad e = 611.2 \exp\left(\frac{17.67\,T_d}{T_d + 243.5}\right),$$

where T_d is in °C and p_0 and e are in Pa. Second, the standard lapse rate is replaced with a location- and season-dependent value from GPT3, an empirical troposphere model that provides a lapse rate for any location and time of year [28]. Values are interpolated bilinearly on the 1° grid. This formulation does not resolve humidity aloft.

All three conversions are implemented in R. Both ERA5 conversions are available in the GeoPressureR function `pressure_to_altitude()` (`altitude_formula = "standard"` or `"virtual"`). ERA5 single-level variables [31] are read from the ECMWF ARCO archive at the nearest 0.25° grid cell and hour.

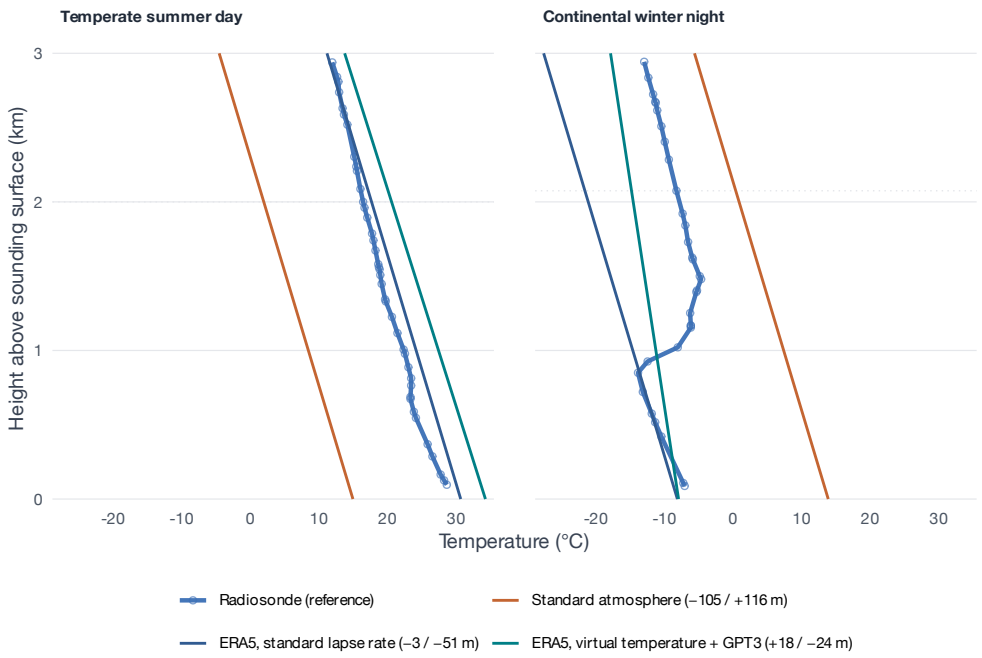


Figure 1: Radiosonde temperatures and the temperature profiles assumed by the three pressure-to-altitude conversions for a temperate summer day (left) and a continental winter night (right). Colours identify each profile; the legend also gives its signed height error near 2 km above the sounding surface (summer / winter), after subtracting the surface error. The radiosonde is the reference profile. The examples were selected because their errors were close to the class medians.

2.2 Reference datasets

We test pressure-to-altitude conversion in two settings: at weather stations with reported elevations, and during balloon ascents, where measured pressure can be compared with the reported height profile. Figure 2 shows the locations used in both comparisons; full data-screening criteria are in Additional file 1: S1.

2.2.1 Ground-level reference: GHCNh

For the ground comparison, we use NOAA's Global Historical Climatology Network hourly (GHCNh) archive [32], which provides hourly surface pressure and station elevation. We select one station per represented 200 km equal-area cell, using its latest qualifying 12-month window within 2022–2026. To ensure the reported elevation is a useful reference, we remove pressure observations outside 300–1100 hPa or marked as suspect by GHCNh, then exclude stations with large persistent pressure-derived elevation offsets from ERA5, abrupt changes or frequent implausible readings. This

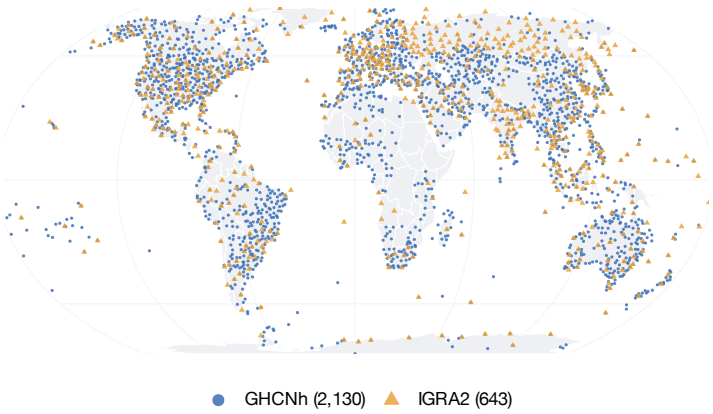


Figure 2: Weather stations and radiosonde launch sites used to evaluate pressure-derived altitude at the surface and through the atmosphere. Station counts are shown in brackets; complementary ground references are described in Additional file 1: S4.

retains 2,130 stations; full thresholds are given in Additional file 1: S1. Each retained pressure report is paired with ERA5 conditions from the nearest hour.

Because listed station elevation may not exactly match pressure-sensor height, we also assess two ground references with more explicit height information: pressure sensors at GNSS stations with checked site-log metadata and MeteoSwiss stations with published pressure-sensor heights. These comparisons test how reference-height information affects the results and are presented in Additional file 1: S4.

2.2.2 Flight-height reference: IGRA2

To assess how conversion error changes above the surface, we use balloon soundings from NOAA's Integrated Global Radiosonde Archive, version 2 (IGRA2) [33–35]. Radiosondes measure pressure as they rise and report the corresponding atmospheric height. We compare pressure-derived height changes with reported height changes, from the surface to 6 km above it, using 643 stations and the latest qualifying 12-month windows within 2022–2026. Each profile level is paired with ERA5 conditions from the nearest hour.

2.3 Error metrics

We treat reference pressures as exact conversion inputs and define error as estimated minus reference height. We use the same summaries on the ground and in flight. For **individual observations**, the median absolute error describes a typical error magnitude; the 95th percentile of absolute error describes the upper tail. For **stations**, bias is the mean signed error and SD describes variability around that mean. We

report weighted medians across stations: absolute station bias, |bias|, on the ground, and signed station bias in flight to retain the direction of height error. Median |bias| describes a persistent station offset; it differs from the median absolute error of individual observations.

On the ground, error is converted altitude minus the reported reference elevation. Individual hourly errors describe single reports; station bias and SD separate persistent offsets from temporal variability. Each retained GHCNh station receives equal weight, divided among its accepted hourly observations; GNSS and MeteoSwiss stations receive equal weight. All three conversions use the same records and weights. Because the GHCNh plausibility screen uses ERA5 residuals, we compare it with coverage-only selection within the same represented cells (Additional file 1: S5). Terrain, season, local time and climate are assessed in Additional file 1: S2.

In flight, error is the converted height difference from the sounding surface minus the corresponding radiosonde height difference. Individual absolute errors are summarized within height bands. Station bias and SD are calculated across retained levels and soundings within each station–height band, separating persistent offsets from variation among profiles and levels. Figure 4 shows their weighted medians and station interquartile ranges in 250 m bands. Within each reporting band, contributing stations receive equal weight, divided equally among their retained levels.

For the overall bird summary, we use the same median absolute error, with each height band's error distribution weighted by its share of recorded flight time. These shares are estimated from 203 tagged birds of 25 species in the Master GeoLocator Data Package [36], accounting for the sampling interval represented by each record (Figure 4 c). The weighted median is calculated from the combined individual-error distribution, rather than by averaging band medians. It describes this sample's height distribution, not an individually validated bird flight. The construction of these weights and an equal-tag comparison are described in Additional file 1: S6.

3 Results

3.1 Ground altitude errors

Across 2,130 screened GHCNh stations, the median / 95th percentile of absolute hourly errors fell from 40.3 / 148 m with the standard atmosphere to 4.1 / 23 m with ERA5 and the standard lapse rate (Figure 3). These pooled errors describe individual reports; the station summaries separate persistent offsets from temporal variability. The weighted median station |bias| and SD fell from 26.0 and 52.1 m, respectively, to 3.1 and 3.4 m with ERA5. Adding virtual temperature and the GPT3 lapse rate made little difference to ground errors.

Station offsets and variability were larger in rougher terrain: median |bias| rose from 2.6 m in flat terrain (sub-grid ERA5 orography SD below 20 m) to 6.8 m in mountains (above 300 m); station SD rose from 3.0 to 8.1 m. The ERA5 median absolute hourly

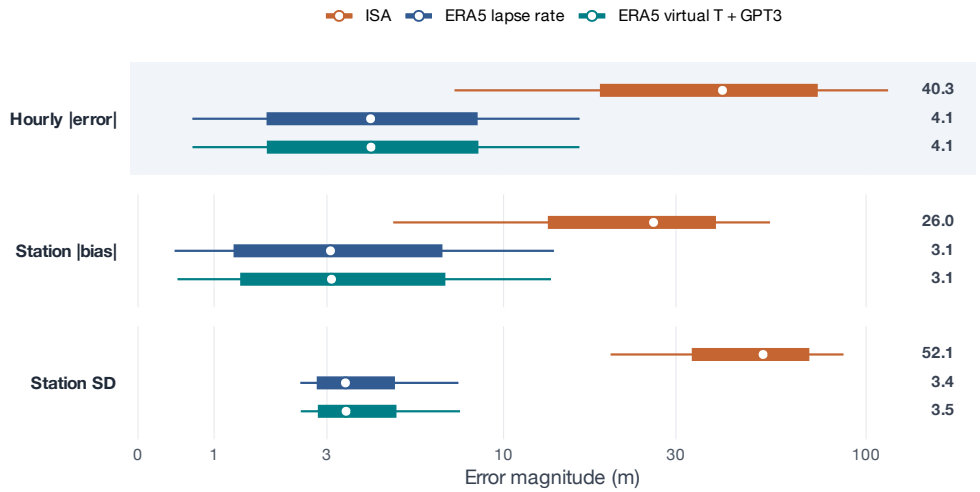


Figure 3: Ground errors at screened GHCNh stations. The lightly shaded top panel shows the weighted distribution of absolute errors across hourly observations. The lower panels group station-level absolute bias and within-station SD, separating persistent offsets from temporal variability. Thick segments show the weighted 25th–75th percentiles, thin segments the 10th–90th percentiles, and white dots mark weighted medians; numbers at the right give those medians in metres. All panels share a pseudo-log x-axis (linear near zero and logarithmic for larger errors; sigma = 1 m); each retained station contributes equal total weight.

error was similar with the references that provide more explicit pressure-sensor heights: 3.0 m at 49 GNSS stations and 4.2 m at 138 MeteoSwiss stations, compared with 4.1 m at GHCNh (Additional file 1: S4). At 3 matched Swiss stations, substituting published pressure-sensor height for listed GHCNh elevation reduced median |bias| from 1.6 to 1.4 m. This indicates that height metadata can account for part of the remaining offset at these sites, without establishing the cause of discrepancies at other stations (Additional file 1: S4).

3.2 Flight-height errors

At individual profile levels, median absolute error at 2–3 km above the sounding surface was 93.4 m with the standard atmosphere, 32.2 m with ERA5 and the standard lapse rate, and 24.5 m with virtual temperature and GPT3. The corresponding 95th percentiles of absolute error were 237, 120 and 84 m. At station level, Figure 4 separates signed bias from within-station SD: the corrected conversion brings the median station bias towards zero, while variation among profiles and levels remains. At 2–3 km, adding virtual temperature and GPT3 shifted median station bias from -33.2 to -5.5 m, while median station SD remained similar (34.2 and 33.8 m). These station summaries use the full 2–3 km reporting band, while Figure 4 uses finer bands.

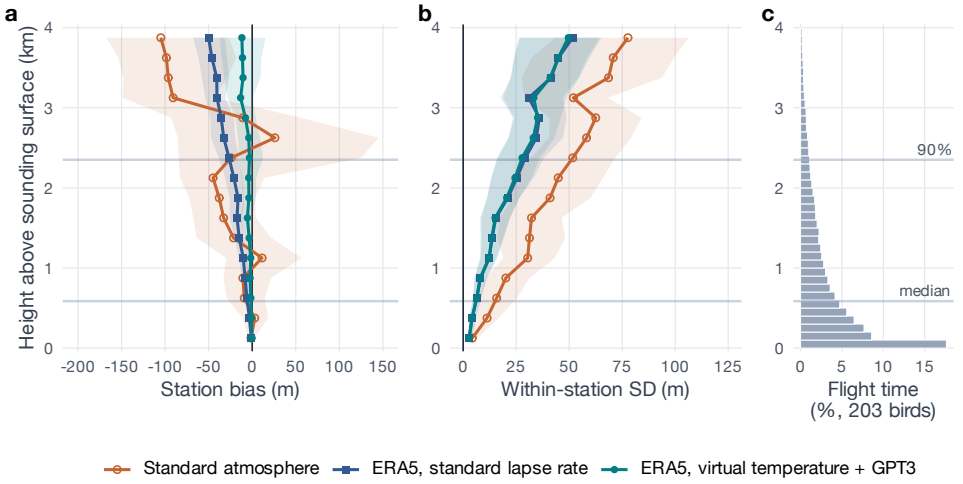


Figure 4: Surface-referenced flight-height errors against height above the sounding surface: (a) station bias (mean error; negative: height underestimated) and (b) station error SD within each height band. Lines and points: equally weighted medians across stations in 250 m height bands; shaded bands: weighted 25th–75th percentiles for all three conversions. Curves join observed-band summaries without smoothing; the highest band is centred at 3.875 km. These station quantiles differ from the pooled individual-error summaries. The display covers 0–4 km; error calculations extend to 6 km. (c) Flight-time height distribution of 203 birds, with its median (590 m) and 90th percentile (2,350 m) marked across the panels.

After weighting profile-level errors by the tracked birds' flight-time distribution across height bands, median absolute error fell from 20.6 to 6.9 and 5.5 m. All these errors are referenced to each sounding's own surface and exclude the absolute ground-reference error.

4 Discussion

Our results support a practical improvement to pressure-derived bird altitudes: combine local ERA5 surface conditions with virtual temperature and a GPT3 lapse rate. Accounting for weather provides the largest gain, reducing median absolute ground error from 40.3 to 4.1 m. The profile refinements provide an additional benefit in flight, bringing systematic height bias towards zero while leaving much of the within-station variability unchanged. Together, these improvements reduce flight-time-weighted median absolute error from 20.6 m with the standard atmosphere to 5.5 m. Both the weather inputs and the lapse-rate climatology are openly available, making the combined conversion a practical option for long tracking records.

Reducing these errors can help distinguish ecological patterns from atmospheric variation. At a fixed ground elevation, weather-driven pressure changes can appear as gradual, continuous elevation changes when converted with a standard atmosphere. Correcting for weather helps separate this apparent movement from the timing and magnitude of actual elevational movements. In flight, the ecological importance of an altitude error depends on the vertical scale of the question: a difference of a few tens of metres may leave a broad description of kilometre-high migration unchanged, yet matter when estimating exposure to turbine rotors [13, 23], responses to urban environments and artificial light [37, 38], or flight relative to topography [39]. For example, a radar experiment found a mean flight-height reduction of about 23 m under white-light treatment [38], illustrating that ecologically relevant responses can occur at this scale. These applications motivate more accurate individual tracking records, although their sensitivity to our correction remains to be evaluated. Atmospheric errors also vary with season and time of day (Additional file 1: S2), so comparisons of altitude among these conditions can inherit conversion differences. The bird-weighted summary describes the sampled distribution of flight time, and giving each tag equal weight preserves the conversion ranking (Additional file 1: S6); height-specific errors remain more relevant to a particular flight or narrow height range.

The atmospheric information needed for correction need not make altitude estimation cumbersome. Weather-station and reanalysis references have already been used in bird studies [3, 6, 26], but fixed standard-atmosphere conversions also remain in use [7, 21, 22]. ERA5 provides hourly atmospheric conditions across the globe [24, 31], allowing weather correction without obtaining local station measurements along a track. GPT3 adds a reusable global climatology of geographic and seasonal lapse-rate variation [28]; virtual temperature requires the additional surface humidity information available from ERA5. GeoPressureR combines these inputs in its pressure-to-altitude workflow [40], so the refinements require little additional user effort once pressure, time and an estimated position are available. High horizontal precision is not a prerequisite for useful atmospheric correction: in our ten-station pilot near 2 km, displacing the query by 50 km changed the corrected absolute-altitude median error by less than 1 m, while the median absolute change in paired estimates was about 5 m (Additional file 1: S7). Individual changes were larger in the tail, and these results concern atmospheric annotation rather than identifying terrain beneath a bird.

The choice of atmospheric inputs nevertheless matters. ERA5-Land's finer grid did not improve ground altitude and produced larger persistent offsets across the three reference networks, whereas surface-referenced flight errors were similar between products (Additional file 1: S3). This pattern points to differences in the surface reference and supports using ERA5 single-level conditions for the evaluated conversion. For the atmospheric column, the virtual-temperature/GPT3 correction reduces recurring bias without requiring a temperature profile for every measurement. Its limited effect on within-station scatter is consistent with a climatology that cannot reproduce each day's conditions, including inversions. Variable lapse rates derived from ERA5 pressure levels have already been used in bird tracking [26], and fuller atmospheric profiles have been evaluated for barometric altitude in aviation [41]. Comparing such

Table 1: Summary of evaluated conversion errors (m). Ground: median absolute station bias in flat terrain and mountains (sub-grid ERA5 orography SD below 20 m and above 300 m), and median within-station SD, at screened GHCNh stations. Flight: median absolute profile-level error relative to the sounding surface; the final row additionally weights the combined error distribution by recorded bird flight time. All values exclude added tag-sensor error; the ground comparison retains reference-height uncertainty.

Quantity	Standard atm.	ERA5 standard	ERA5 virtual + GPT3
Ground, flat terrain (median |bias|)	23.4	2.6	2.6
Ground, mountains (median |bias|)	43.0	6.8	7.4
Within-station scatter (median SD)	52.1	3.4	3.5
Flight, 1–1.5 km (median |error|)	55.4	13.7	9.3
Flight, 2–3 km (median |error|)	93.4	32.2	24.5
Flight-time weighted (median |error|)	20.6	6.9	5.5

approaches with the present conversion would establish when their additional data requirements yield a useful improvement for tracking records.

The remaining uncertainty depends on the quantity being interpreted (Table 1). Persistent ground offsets can partly cancel in comparisons at one site, making relative elevation changes less sensitive to those offsets than absolute elevations, although temporal variability remains. Both offsets and variability increase in rough terrain. Sensor offset and drift add uncertainty beyond the atmospheric conversion; known-height events, such as floating periods in seabirds, can help calibrate records [13, 23]. Such calibration complements weather correction but does not determine the temperature profile aloft. Controlled laser-reference trials [27] and evaluations at grounded birds [26] address complementary parts of the measurement process, so their reported errors cannot be directly ranked against our conversion errors. Local ground clearance additionally requires terrain or sea-surface elevation in a consistent vertical reference [19, 42]. For example, the corrected conversion's absolute-error p95 of 20 m at 100–250 m is relative to the sounding surface; it excludes absolute surface-reference error and added sensor error. It therefore represents only part of the uncertainty relevant to a bird's position relative to a rotor or terrain feature.

The reference comparisons constrain how far these findings transfer to deployed tracking records. Ground errors depend on reference-height information and on selection: the ERA5-based GHCNh checks can exclude genuine model discrepancies as well as suspect records. The coverage-only comparison preserves the conversion ranking, with a larger effect on upper-tail errors than on medians (Additional file 1: S5). The paired Swiss height substitution illustrates a metadata contribution at those sites, while GNSS and MeteoSwiss provide complementary references with more explicit sensor heights (Additional file 1: S4). These comparisons do not establish independence from ERA5 assimilation. Reported radiosonde heights can likewise be derived

from the measured atmospheric profile, so our flight comparison primarily tests the assumed atmospheric column rather than independently measured geometric height. ERA5 assimilates conventional surface and upper-air observations [24]; assimilation of the evaluated records was not determined. Balloon drift, uneven geographic coverage and the absence of offshore references further limit transfer. Within this scope, the results support using readily accessible weather and climatological information to reduce avoidable conversion errors, while selecting uncertainty summaries appropriate to the height, vertical reference and ecological question.

5 Conclusions

A fixed standard atmosphere introduces substantial errors into pressure-derived bird altitudes. At screened ground-reference stations, its median absolute error was 40.3 m and 5% of absolute errors exceeded 148 m. Referencing the conversion to local ERA5 conditions reduced these to 4.1 and 23 m. In flight, the standard-atmosphere 95th percentile of absolute error reached 237 m at 2–3 km above ground, even after referencing each flight height to its own surface. ERA5 greatly reduced these errors, while virtual temperature and a GPT3 lapse rate further reduced flight-time-weighted median absolute error from 6.9 to 5.5 m. These improvements support more reliable interpretation of migration altitude and reduce one source of uncertainty in conservation assessments.

The corrected conversion is computationally simple: it combines surface atmospheric variables with a climatological lookup and is available in a single software function. Users supply measured pressure and obtain atmospheric conditions using an estimated bird position at the measurement time; no measured temperature profile aloft is required. This makes the correction practical for long geolocator records. It substantially reduces conversion error, while day-to-day temperature profiles, unresolved terrain, pressure-sensor errors and position uncertainty still constrain individual altitude estimates, particularly near ecological or turbine-height thresholds.

6 Declarations

Ethics approval and consent to participate. Not applicable: no animals were handled for this study; the bird flight heights come from published tracking data.

Consent for publication. Not applicable.

Availability of data and materials. Code, derived tables, figures and data-archive instructions are available at https://github.com/GeoPressure/altitude-validation [43]. The bird weighting uses the open-access Master GeoLocator Data Package, version 2026.0 [36]; the release-specific DOI and verified input checksums accompany the derived weights. The conversions are implemented in GeoPressureR [40]. GHCNh, IGRA2, GNSS, MeteoSwiss and ERA5 data are openly available from their providers (Additional file 1: S1 and S4).

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. RN designed the study, performed the analysis and wrote the manuscript. MG contributed to the implementation of the conversions in GeoPressureR and GeoPressureAPI and revised the manuscript. [To be confirmed.]

Acknowledgements. [To be completed.]

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